

The Lifelong Criminal Consequences of Kindergarten Cutoff Dates

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Working Paper†

Abstract

I exploit exogenous variation in kindergarten entry cutoff dates based on date of birth and use administrative prison data with a regression discontinuity design (RDD) to measure their effects on adult crime. Kindergarten entry cutoffs create quasi-random variation in school starting age, with children born just after the cutoff delayed entry by one year, reducing required schooling under age-based compulsory schooling laws. I draw on Florida and Illinois, two states that gradually moved their cutoff dates earlier by one month each year during the 1980s. Exploiting changes in school entry cutoffs, I link birth date to lifelong incarceration outcomes and find that delayed entry increases crime among White females by at least 16%, consistent with literature on higher returns to education for women. Using birth certificate and ACS data, I show this effect reflects an increase in the crime rate rather than changes in births or migration across the cutoff.

JEL: I21, K42, J24

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I Introduction

A large body of literature connects human capital accumulation to criminal behavior, both theoretically (e.g., Becker 1968; Ehrlich 1973; Lochner 2004) and empirically (e.g., Angrist and Krueger 1991; Lochner and Moretti 2004; Anderson 2014; Baron, Hyman, and Vasquez 2024). Education raises the opportunity cost of crime, making criminal activity less appealing and shifting the margin for individuals choosing between legal and illegal behavior. The size of this effect depends on the wage gains associated with higher schooling, which the literature finds to be larger for women than for men (Lochner and Moretti 2004; Dougherty 2005).

In the U.S., required education is determined at the state level through compulsory schooling laws that mandate school attendance for specified ages.¹ Several papers exploit compulsory schooling laws to instrument for education and measure its effects on adult earnings (e.g., Angrist and Krueger 1991; Oreopoulos 2006; Brunello, Fort, and Weber 2009), arrests and incarcerations (e.g., Lochner and Moretti 2004; Hjalmarsson, Holmlund, and Lindquist 2015; Bell, Costa, and Machin 2022), and other outcomes (e.g., Lleras-Muney 2005; Clark and Royer 2013).

Related work measures the effect of beginning school later due to kindergarten entry cutoff dates, which determine when a child is eligible to start school. Individuals born on each side of the cutoff are, on average, similar in all ways aside from when they start school. Those born before the cutoff start school at nearly five, while those born afterward delay entry by a year and start when they are almost six. These papers generally find delayed school entry increases early test scores which fade with time (Elder and Lubotsky 2009; Lubotsky and Kaestner 2016), decrease educational attainment (Dobkin and Ferreira 2010; Kaplan, Spenkuch, and Tuttle 2025), and decrease earnings through age thirty (Black, Devereux, and Salvanes 2011).² Fewer papers estimate the criminal consequences of kindergarten entry cutoff dates.

Related papers consistently find that delayed individuals have less juvenile crime due to incapacitation (Cook and Kang 2016; Depew and Eren 2016; Shepard 2024). Using a regression discontinuity design (RDD), Cook and Kang (2016) find delayed entry to increase crime among nineteen year olds in the US. Using a similar design in Denmark, Landersø, Nielsen, and Simonsen (2015) find delayed entry to decrease young adult crime, as compulsory schooling laws there are attainment-based rather than age-based. Attainment-based compulsory schooling laws, prevent the discontinuity of required education based on school entry which occurs with age-based compulsory schooling laws common in the U.S. Only McAdams (2016) estimates the effect of delayed entry from the cutoff on adult crime. He does so using a difference-in-differences model comparing

¹Failure to comply with compulsory schooling laws may result in fines and criminal prosecution of the parents possibly resulting in a misdemeanor and jail time (U.S. Department of Education 2024; Florida Legislature 2025).

²Rossin-Slater, Sabety, and Wu (2026) and McCrary and Royer (2011) find delayed school entry to decrease the probability of a learning disorder diagnosis in pre-school as well as no effect on childbearing, respectively.

those born in the quarter before the cutoff to those born in the quarter after. Using the 1970 and 1980 decennial Censuses he finds delayed entry to increase individuals' crime rate by 3%. However, a limitation shared by prior work is the use of longstanding cutoff dates which allow parents to sort around cutoffs in ways which may not be observable, given evidence of strategic birth timing (Buckles and Hungerman 2013; Clarke, Oreffice, and Quintana-Domeque 2019). My paper builds on Cook and Kang (2016) and McAdams (2016) by being the first to exploit changes in kindergarten entry cutoff dates after conception, ruling out parental planning, and estimating the effect on lifelong crime.

My setting is Florida and Illinois where each moved their kindergarten entry cutoffs earlier in the school year during the 1980s while leaving compulsory schooling laws unchanged. The cutoff dates therefore provide the sole source of variation in required schooling among compliers, creating a discontinuity where the running variable is an individual's date of birth. I use administrative prison population data from each state, which include exact dates of birth, and a RDD to measure the effect that delayed school start has on an individual's lifetime criminal history. Unique to my paper are the changing kindergarten entry cutoff dates, which occurred after conception. Such changes reduce potential manipulation around the cutoff due to parents timing their child's birth since their child was already conceived by the time of announcement.

Consistent with larger returns to high school for women, I find delayed entry causes more imprisonments among females, with this effect concentrated among White women where both labor market and marriage market returns to schooling are largest. I observe no effect for males. I also find births, parental characteristics, and migration to be smooth across the cutoff. Together my results imply the increase in female imprisonments is due to an increase in the crime rate, rather than the number of births.³ In total I estimate White women born just after the cutoff have at least a 16% higher crime rate relative to those born just before the cutoff.

II Institution

School entry cutoff dates refer to the date used to determine whether a child starts school in the current year or must wait until the following year.⁴ In both Florida and Illinois, children must turn five on or before the cutoff date of the school year to be eligible to begin kindergarten that same school year. Prior to 1979, Florida had a kindergarten birthday cutoff of January 1 and Illinois had a cutoff of December 1.

Neither state had changed the cutoff date since its inception in 1965 and 1883 for Florida and Illinois, respectively (Abbott and Breckinridge 1917; Florida House of Representatives 1999). In June 1979 Florida

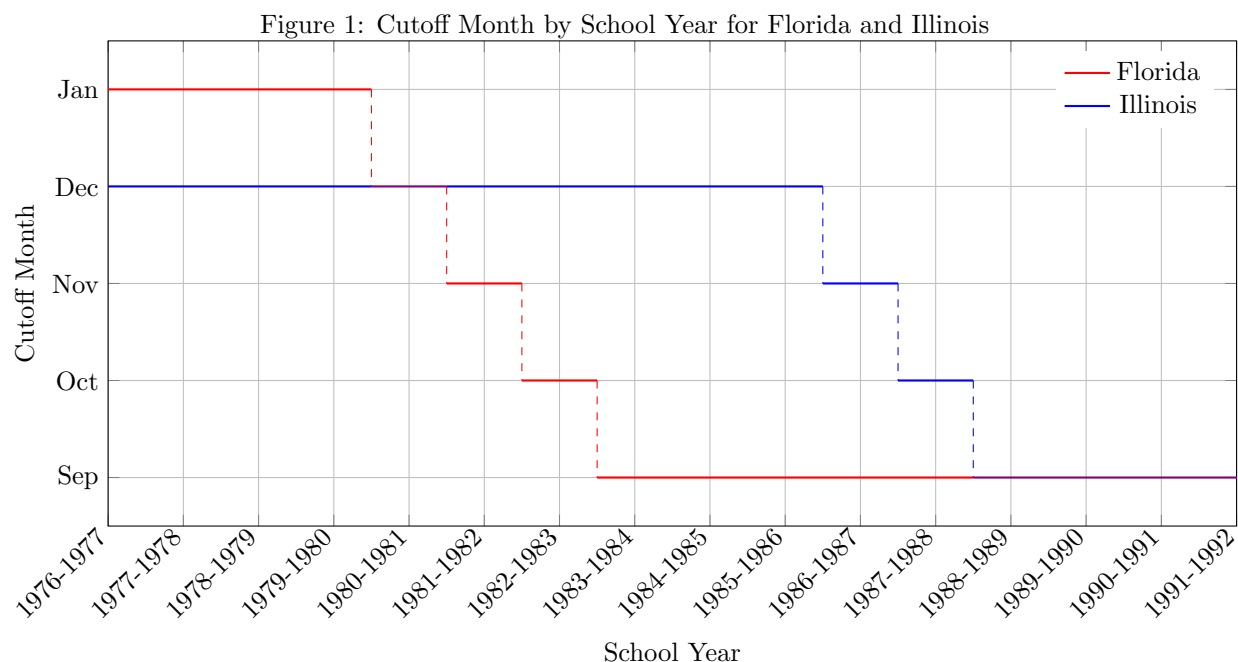
³Assuming that the probability of arrest conditional on committing a crime and the probability of imprisonment conditional on arrest do not change discontinuously at the cutoff, as is common in the literature.

⁴I use kindergarten and school entry cutoffs synonymously.

enacted the change in the cutoff from January 1 to September 1 over four years (Florida Legislature 1979). This initially moved the kindergarten cutoff date to December 1, rather than January 1, during the 1980–1981 school year and delayed the school entry for children born between December 2, 1975 and January 1, 1976 (inclusive). Each subsequent school year had a cutoff one month earlier than the previous one until the cutoff remained at September 1 beginning in the 1983–1984 school year (Whaley 1985).

Illinois had a similar change in its kindergarten cutoff caused by amendments to the Illinois School Code (105-ILCS 5) in September 1985 (Education Week 1987). This moved the cutoff by one month per year for three years. This first affected the cohort beginning kindergarten in the 1986–1987 school year, for whom the cutoff date moved from December 1 to November 1. This delayed school entry for children born between November 2, 1981 and December 1, 1981 (inclusive).

Due to changes in the cutoffs, bias due to parental planning may only occur if parents knew of the change in cutoffs up to five years in advance and were aware of the cutoff change schedule. Once the cutoff date was moved to September 1, it remained unchanged and continues to serve as the kindergarten entry cutoff in both Florida and Illinois (Education Commission of the States 2014). I include a timeline of the change in cutoff dates for Florida and Illinois in Figure 1 and a more detailed version in Table 4 of the Appendix.

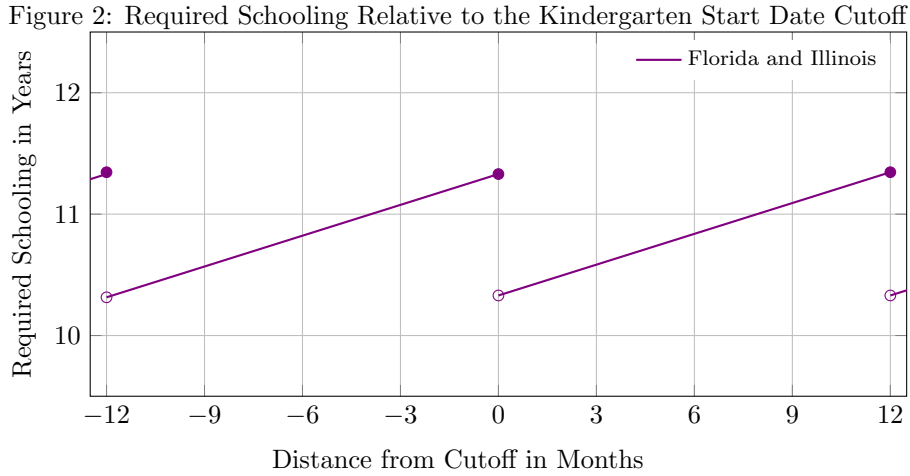


Note: The school cutoff was always the first day of the month. Florida first implemented the reform in the 1980–1981 school year while Illinois did so in the 1986–1987 school year (Florida Legislature 1979; Education Week 1987; Florida House of Representatives 1999; Florida Senate Committee on Education Innovation 1999). Changes in the cutoff date were announced in June 1979 in Florida and September 1985 in Illinois. A more detailed version is in Table 4 in the Appendix.

Importantly, neither Florida nor Illinois changed their compulsory schooling laws during this period of

changing kindergarten cutoff dates (Florida Legislature 1970; Illinois General Assembly 1970; Illinois State Board of Education 1983; Angrist and Krueger 1991; Knapp et al. 2025). Students in Florida and Illinois could not drop out until age sixteen and required a parent or guardian’s permission.⁵ Upon turning eighteen in either state, individuals can leave school without permission from a parent or guardian. Thus, all variation in required schooling for those affected by changing cutoff dates is determined solely by the cutoff date itself.

The combination of kindergarten entry cutoffs and compulsory schooling laws generate a discontinuity in required schooling based on a child’s date of birth. Those born just before the cutoff have over eleven years of required schooling while those born just after have only ten years. Figure 2 illustrates this discontinuity, which arises solely from kindergarten entry cutoffs.



Note: Required schooling is calculated as the difference in age between the age at which a child begins kindergarten, determined by the cutoff dates, and the earliest age at which a child can opt out. Students could opt out of schooling at the age of sixteen with a parent or guardian’s permission during this period in Florida and Illinois (Florida Legislature 1970; Illinois General Assembly 1970; Illinois State Board of Education 1983; Angrist and Krueger 1991; Knapp et al. 2025).

III Theory

I model my setting using a simple two period framework that links kindergarten entry cutoffs to later criminal behavior. This model is based on Becker (1968) and Lochner (2004), in which heterogeneous individuals receive schooling in the first period and choose between legal and illegal work in the second period. Their choice of labor supply is determined by the expected payoff of crime relative to their certain payoff of working.

Each individual i has schooling $S_i = s(D_i)$, where D_i is an indicator for those born after the kindergarten entry cutoff date. The function s is weakly decreasing in D_i through the reduction in required education

⁵Illinois has since moved the minimum dropout age to 17 for those born after August 31, 1989.

under age-based compulsory schooling laws. Every individual has human capital $h_i = g_i(S_i, Z_i)$, where Z_i represents all other factors that affect human capital.⁶ The function g_i is strictly increasing in schooling. An individual's human capital determines their legal wage offer $w_i(h_i)$, which is strictly increasing in their human capital h_i .

Every individual i has the opportunity to work either in the legal labor market or the illegal labor market. If an individual chooses the illegal labor market (i.e., crime), they receive a benefit B_i at the risk of apprehension and punishment.⁷ All individuals have an equal probability of apprehension $p \in (0, 1)$ and punishment $F_i(h_i)$ is an increasing function of human capital due to rising opportunity costs from legal work, $w_i(h_i)$. This results in the total payoff from crime, π_i^C , which depends on the benefit and the expected punishment. Individuals choose to commit crime if their payoff, π_i^C , is strictly greater than the payoff from legal work, $w_i(h_i)$.⁸ When the individual chooses crime, their decision rule, $c_i(h_i)$, equals one.

Individuals' level of schooling is partially determined by their date of birth relative to the kindergarten entry cutoff. Those born before the cutoff ($D_i = 0$) receive their high level of schooling, $S_i = S_i^H$, for certain, whereas those born after ($D_i = 1$) may receive low schooling, S_i^L , with probability q_i where $q_i \in (0, 1)$. Therefore, the cutoff alone changes the distribution of schooling toward lower attainment for delayed individuals than they would have had if they were born before the cutoff, consistent with the literature (e.g., Dobkin and Ferreira 2010; Cook and Kang 2016; Kaplan, Spenkuch, and Tuttle 2025).

$$Pr(S_i = S_i^L | D_i = 1) = q_i, Pr(S_i = S_i^H | D_i = 1) = 1 - q_i, q_i \in (0, 1) \quad (1)$$

Since the decision rule is determined by the realization of S_i and schooling only affects the legal wage offer through human capital, an individual who commits crime when they have high schooling, S_i^H , would have chosen to commit crime if they had low schooling, S_i^L . This makes the probability of committing crime with low human capital, $c_i(h_i^L)$, no less than with high human capital, $c_i(h_i^H)$.

$$c_i(h_i^H) = 1 \Rightarrow c_i(h_i^L) = 1 \Rightarrow c_i(h_i^L) \geq c_i(h_i^H) \quad (2)$$

⁶Two other mechanisms may be added to the model without changing its prediction under modest assumptions. First, one's age relative to peers can be included through the human capital function, so long as the required schooling effect dominates the effect of relative age, as supported by empirical literature (Dobkin and Ferreira 2010; Kaplan, Spenkuch, and Tuttle 2025). Second, delayed individuals spend an extra year at home before beginning kindergarten, given the near-absence of public pre-K in Florida and Illinois and low national enrollment during this period (National Center for Education Statistics 2016). This effect can be incorporated into the human capital function without changing the model's predictions if the extra year at home weakly reduces human capital, consistent with Cascio (2009).

⁷I assume the benefit of crime B_i is independent of schooling and human capital, consistent with the literature (e.g., Becker 1968; Ehrlich 1973; Lochner 2004). This assumption can be relaxed without altering equation 3. It is sufficient that additional schooling raises legal earnings more than criminal returns, that is $\frac{dB_i(h_i)}{dS_i} \leq \frac{dw_i(h_i)}{dS_i}$ for all S_i .

⁸I assume individuals are risk averse at the inflection point.

The probability of crime for those born before the cutoff is equal to $c_i(h_i^H)$.⁹ The probability of crime for individuals born after the cutoff is a convex combination of the two decision rules, $c_i(h_i^L)$ and $c_i(h_i^H)$, weighted by the probability of low education, $q_i \in (0, 1)$.¹⁰ It follows that the probability of crime for those born after the cutoff is always at least as high as it would have been if they were born before the cutoff, shown in equation 3.

$$Pr(c_i(h_i) = 1 | D_i = 0) \leq Pr(c_i(h_i) = 1 | D_i = 1) \quad (3)$$

In total, this simple model demonstrates how cutoff-induced delays in school entry lower schooling, thus reducing human capital and wages and increasing the likelihood of criminal activity. The size of the effect on crime depends on both the probability of reduced schooling, q_i , and the wage gap between high and low schooling outcomes, $w_i(h_i^H) - w_i(h_i^L)$. Because the crime effect scales with the wage gap, the model predicts larger effects for individuals with larger individual and household returns to education such as women (e.g., Behrman, Rosenzweig, and Taubman 1994; Card 1999; Dougherty 2005; Chiappori, Iyigun, and Weiss 2009). Furthermore, because the model operates through legal wages and opportunity costs, it predicts effects to be concentrated among financially motivated crimes, measured by property crimes (Gould, Weinberg, and Mustard 2002; Lochner and Moretti 2004). Overall, kindergarten entry cutoffs quasi-randomly affect human capital and wages which manifest in long run crime outcomes that I estimate using a regression discontinuity design.¹¹

IV Data

I use public data from the Florida Offender Based Information System (FOBIS) and the Illinois Department of Corrections (IDOC). FOBIS includes all Florida prisoners admitted since 1981, while IDOC includes all prisoners in Illinois since 2005. Because the initially affected Illinois cohort turned twenty-four in 2005, I do not observe prison stays during their early adult years, which attenuates my estimated effect in Illinois.¹² Both datasets include the date of birth for all prisoners admitted, so eligibility for kindergarten is known.¹³ Other information is available, including race, gender, and offense type, which I use to measure

⁹This is because there is no uncertainty in the amount of human capital. An individual's probability of crime is 1 when their expected benefit from crime, $\pi_i(h_i^H)$, is greater than their wage, $w_i(h_i^H)$, and is zero otherwise.

¹⁰The uncertainty in crime for those born after the cutoff comes from the random assignment of schooling, not from any uncertainty in individuals' decisions in the second period.

¹¹Additionally, the framework can be extended to allow individuals to choose between misdemeanor and felony offenses. Because my outcome is imprisonment, a sanction rarely applied to misdemeanors, any shift from misdemeanors to felonies is empirically indistinguishable from a shift from nonoffending into felony crime. This extension can be made by allowing the punishment to be increasing in the severity of the crime such that the expected payoff of felony offenses falls more quickly with education than the payoff of misdemeanors.

¹²Table 5 of the Appendix details the years of criminal history observed for each cohort.

¹³I make the implicit assumption that all prisoners observed grew up in the same state they went to prison in. Similar assumptions are made in the literature (Cook and Kang 2016; McAdams 2016) and will attenuate my results since I do not have data on where prisoners were born nor when and where they started school.

heterogeneous effects.

Both FOBIS and IDOC data are reported at the prison-stay-level. I convert each of these datasets to be at the date of birth level, with each date of birth including the number of unique prisoners born on that day by demographics and crime type.¹⁴ Each date of birth is then assigned a distance to the nearest kindergarten entry cutoff. I restrict the sample to cohorts affected by the cutoff changes who were conceived before the announcement to reduce potential selection. This restriction leaves me with five cohorts per state. Summary stats for the entire cohorts and for those within a ten day bandwidth of the cutoffs are included in Table 6 of the Appendix.

I use natality data from the National Vital Statistics System (NVSS) of the National Center for Health Statistics and the American Community Survey (ACS) to test whether observed factors are smooth around the cutoff dates. NVSS data include child birth characteristics such as the number of births, race, prenatal care, and birth weight. Other variables include the race, age, and education of the mother and (sometimes) father on all births in the affected cohorts in Florida and Illinois.¹⁵ These data are at the child-level, which I convert to the date of birth level. ACS data remain at the individual level, but the date of birth is reported only by the quarter of birth. ACS data report state of birth and current state of residence at the time of survey. Using these data, I see estimate migration into or out of either Florida or Illinois. I provide more detail on how the NVSS and ACS datasets are used for validity checks in Section V.B.

V Identification Strategy

V.A Adult Crime

Children in Florida and Illinois must turn five on or before the cutoff date of the school year in which they begin kindergarten. This creates a discontinuity where the running variable is the distance to the cutoff, which is perfectly determined by an individual's date of birth and state of residence. Children born on September 1 and September 2 are only one day apart, but the older child begins school a year earlier. Additionally, because compulsory schooling laws are age-based, they become eligible to leave school only one day apart. Thus, the cutoff alone causes those born after September 1 to receive one year less of required schooling.

I exploit the discontinuity at each new cutoff within five years because these children were conceived before the policy change was announced. Assuming that birthdays are as good as random around the

¹⁴I define imprisonment as ever appearing in the data, using an individual's first observed prison spell.

¹⁵Generally, data on the father is only included in the NVSS when the father is either (i) married to the mother of the child and present at the birth or (ii) present at the birth and the father voluntarily provides information.

new cutoffs and there are no confounders, a sharp regression discontinuity design (RDD) gives an unbiased estimate of the intent-to-treat effect of delayed entry on lifetime crime.¹⁶ I pool observations across the first five cohorts affected by the policy change in each state.¹⁷ I use the RDD in equation 4, where my preferred specification uses a triangular kernel and a ten day bandwidth.

$$Y_{dts} = \beta_0 + \beta_1 \text{Delayed}_{dts} + \beta_2 f(\text{DateOfBirth}_{dt} - \text{Cutoff}_{dts}) + \beta_3 \text{Delayed}_{dts} \times f(\text{DateOfBirth}_{dt} - \text{Cutoff}_{dts}) + \mu_s + \mu_t + \mu_{dt} + \varepsilon_{dts} \quad (4)$$

Here, Y_{dts} is the number of individuals born on date d in year t in state s whom I ever observe in prison. Delayed_{dts} is an indicator equal to one if individuals born on day d , year t , and in state s had their kindergarten entry delayed by a year due to the change in cutoff after conception. If Delayed_{dts} equals one, then individuals born on that day would have been subject to the original cutoff prior to the policy change, but are now subject to the new cutoff.¹⁸ $\text{DateOfBirth}_{dt} - \text{Cutoff}_{dts}$ represents the running variable, which is the relative distance between the individual's birthday and their kindergarten entry cutoff date. Unobservable differences, different population sizes, and different birth rates between Florida and Illinois create level differences, so I include state fixed effects, μ_s , to account for time invariant differences between states. As previously mentioned, cohorts' crime in Illinois is observed differentially due to data limitations. I include year-of-birth fixed effects, μ_t , since cohorts have up to ten more years to commit crimes than later cohorts. I include day-of-week fixed effects, μ_{dt} , to account for systematic differences in birth outcomes across weekdays and weekends. Standard errors are clustered at the month-day level.

A statistically significant β_1 indicates a discontinuous change in incarcerations among individuals born just after the cutoff compared to those born before. A positive estimate points to a higher likelihood of imprisonment among those born after the cutoff. Given continuity of births around the cutoff, a positive estimate corresponds to an increase in the crime rate. Thus, receiving up to one year less of education increased the crime rate for those born after the cutoff date.

β_1 captures the intent to treat (ITT) effect of being assigned a delayed kindergarten start based on the date of birth and state of schooling, proxied by the state of incarceration. The ITT is identified under the assumption that births are as-good-as-random around the cutoff. β_1 would be the local average treatment effect (LATE) using a fuzzy RDD if I observed state of birth and the realized school start date for all prisoners

¹⁶These assumptions are common in the literature (e.g., Angrist and Krueger 1991; Cook and Kang 2016).

¹⁷The Illinois cohort born in 1983 is omitted for any results using a bandwidth of more than twenty-nine days. This is because individuals born after September 30, 1983 had increased Medicaid eligibility as studied in Arenberg, Neller, and Stripling (2024). A bandwidth of thirty or more leads to this confounding my identification in the window around the September 1, 1988 cutoff in Illinois.

¹⁸This is not true for those in the initially affected cohorts who have a birth day more than a month after the cutoff. These observations are not included for my preferred specification where the bandwidth is only ten days.

as well as a monotonicity assumption.¹⁹

Lastly, I estimate equation 4 using the individual’s age of first conviction resulting in prison time as an alternative outcome, constructed as the difference between one’s date of birth and date of conviction, which are reported in both datasets. I also examine heterogeneous effects of delayed entry across violent, drug, property, sex, and other crime types. All independent variables remain unchanged for these outcomes.

V.B Validity Checks

V.B.1 Births

Random assignment of births around a given date is a common assumption in empirical work (e.g., Bédard and Dhuey 2006; Black, Devereux, and Salvanes 2011; McCrary and Royer 2011). However, I am the first to use variation in changing kindergarten entry cutoff dates as a source of exogenous variation. The timing of the changes in kindergarten entry cutoffs in Florida and Illinois during this period makes a stronger case for random assignment than if the cutoffs had remained constant. Additionally, Barr, Eggleston, and Smith (2022) find confoundedness with January 1 cutoffs, used in Florida prior to the change, due to differences in public assistance at early ages.

In both Florida and Illinois, these cutoffs changed after the affected cohorts were born or conceived, making it impossible for parents to plan around new kindergarten entry cutoffs. Here, I test whether birth observables are smooth around the cutoff. I use Equation 5 and NVSS data to test this assumption.

$$X_{dts} = \alpha_0 + \alpha_1 Delayed_{dts} + \alpha_2 g(DateOfBirth_{dt} - Cutoff_{dts}) + \alpha_3 Delayed_{dts} \times g(DateOfBirth_{dt} - Cutoff_{dts}) + \mu_s + \mu_t + \mu_{dt} + \nu_{dts} \quad (5)$$

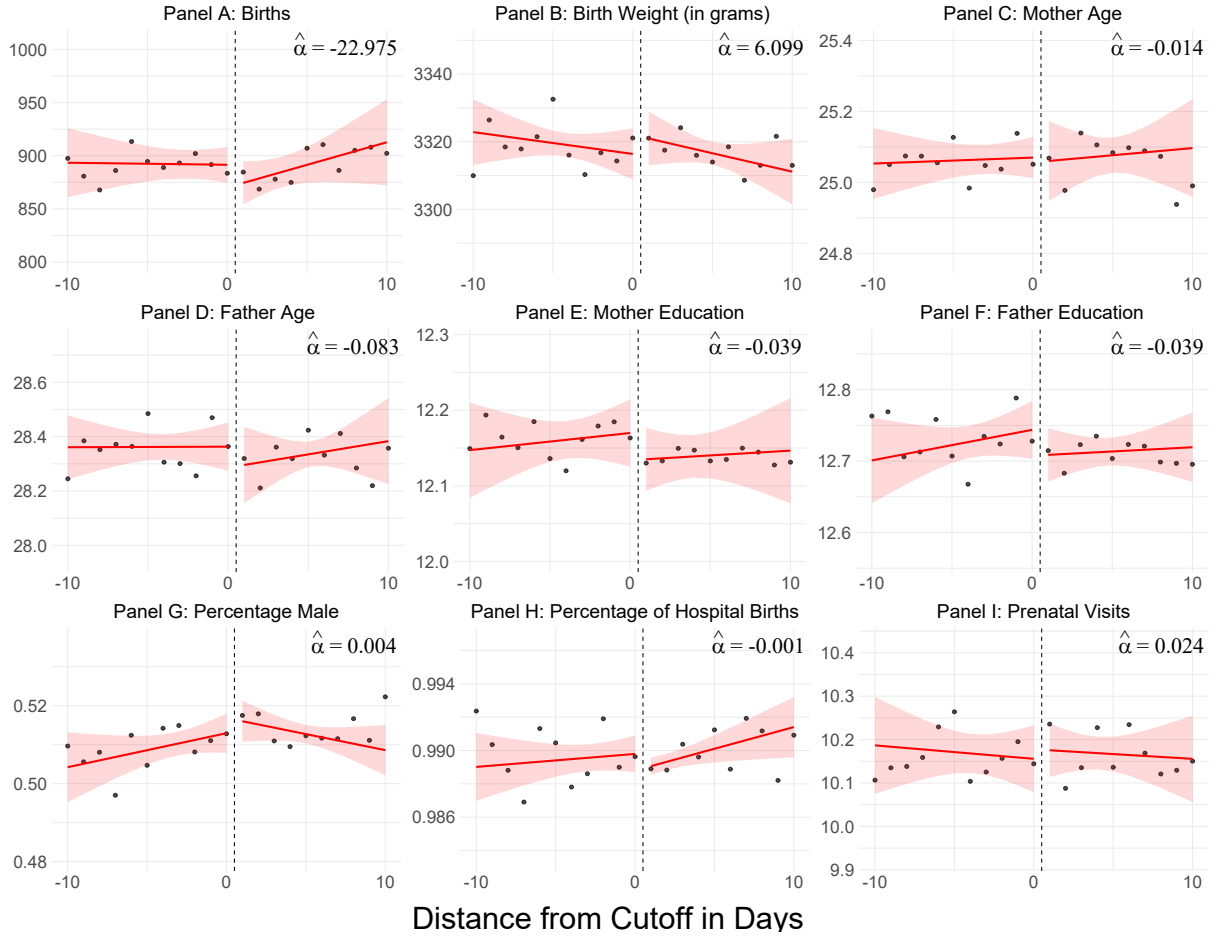
Here, X_{dts} is one of many observables at the time of birth on day d of year t in state s . $Delayed_{dts}$, $Cutoff_{dts}$, μ_s , μ_t , and μ_{dt} are the same as in equation 4. $DateOfBirth_{dt} - Cutoff_{ds}$ is again the running variable, being the relative distance between birthday and the nearest kindergarten entry cutoff date. Standard errors are again clustered at the month-day level. Due to limitations in the NVSS’s demographic variables, I am only able to check smoothness for Whites and Blacks. This covers nearly ninety percent of the prisoner population in Florida and Illinois.

I display the results of Equation 5 where X_{dts} is either the number of births, birth weight, mother characteristics, father characteristics, percentage male, percent born in hospitals, or prenatal visits in Figure 3 and report the results in Table 7 of the Appendix. I find no discontinuity in any observables at birth, lending

¹⁹The monotonicity assumption being that no parents who choose to delay school entry if their child is born before the cutoff would petition for their child to start early if they were born after the cutoff.

credence to my RDD strategy.

Figure 3: Natality Regression Discontinuity Plots



Note: The horizontal axis is the number of days to the kindergarten entry cutoff and is normalized to zero. Individuals with a positive distance were delayed school entry due to the policy change. Zero corresponds to either December 1, 1975, November 1, 1976, October 1, 1977, September 1, 1978, or September 1, 1979 in Florida and either November 1, 1981, October 1, 1982, September 1, 1983, September 1, 1984, or September 1, 1985 in Illinois. Points show residualized bin means after partialling out day-of-week and birth year fixed effects via the Frisch-Waugh theorem (Lovell 1963). The shaded area represents 95 percent point-wise confidence intervals around the fitted regression function, based on heteroskedasticity robust standard errors clustered at the month-day level. Parents' education does not include the initial Florida cohort as it was not reported until 1976. RD results for each outcome are displayed in Table 7 of the Appendix. Source: National Center for Health Statistics (1975-1985).

V.B.2 Moving

Since I don't observe current residence, a potential concern is individuals moving at differential rates which are correlated with their date of birth relative to the cutoff. The ACS includes state of birth, state of residence when surveyed, and quarter of year of birth.²⁰ The month by month change of each policy resulted in an October 1 cutoff for one year in both Florida and Illinois. This coincides with the quarter of birth for that year, 1977 in Florida and 1982 in Illinois, so I compare moving rates of individuals in the third and

²⁰No version of the ACS has more specific dates of birth than quarter.

fourth quarters of those years.

I use the model in equation 6 to test for differences in moving between each side of the October 1 cutoff in Florida and Illinois.²¹

$$\text{Move}_{isq} = \gamma_0 + \gamma_1 \text{Delayed}_{isq} + Z'_i \delta + \mu_s + \eta_{isq} \quad (6)$$

Move_{isq} is an indicator equal to one if individual i currently resides in a state different from where they were born. This definition of Move_{isq} represents either moving into Florida and Illinois or moving out of Florida and Illinois depending on sample restrictions. Delayed_{isq} is similar to previous models where it is equal to one if the individual's quarter of birth was delayed in either Florida or Illinois of that year, otherwise it is zero.²² Z_i is a vector of demographic controls including age, gender, race, and education. μ_s is a series of state fixed effects for the state the individual was born in and standard errors are clustered at the household level.

Equation 6 is the simplest version of my moving model, which I generalize to test for differences in moving among all gender-race groups by interacting Delayed_{isq} with gender, race, and both gender and race simultaneously. Columns 1-4 of Table 8 in the Appendix restrict the sample to those born in Florida and Illinois, giving the dependent variable as an indicator for moving out, while columns 5-8 restrict the sample to those residing in Florida and Illinois making the dependent variable an indicator for moving in.

Overall, I find no evidence of differences in moving habits for all demographic groups other than White males. White males born after the kindergarten entry cutoff move out of Florida and Illinois 3.3 percentage points less than those born before the cutoff. This represents about a 7% decrease in moving among White men relative to the mean. This biases my estimate of adult crime upwards since the White men delayed kindergarten entry are more likely to remain in their respective state and individuals are most likely to commit crime within their state of residence.²³

V.B.3 Placebo

To further establish that my results reflect policy-induced variation in kindergarten entry, I test for an effect at a placebo cutoff of May 1 using the same cohort-years and states as my main analysis. I choose May 1 as it is equidistant from the nearest true cutoffs, September 1 and January 1, minimizing the possibility that the placebo window overlaps with any true cutoff in my sample. No cohort in my sample was ever

²¹The estimates of this model are biased by those born on October 1 although this represents only 1% of the total treated population.

²²In practice, this is an indicator equal to one if the person was born in the fourth quarter of the year (i.e. after September 30) due to the restrictions made on the sample.

²³Approximately 90% of crimes are committed within a 20 minute drive of the criminal's home (Kirchmaier 2024)

subject to a kindergarten entry cutoff near May 1, so any discontinuity indicates that my design is picking up noise, seasonality, or other confounders rather than the policy change itself.

I estimate Equation 4 replacing the true cutoff dates with May 1 of the same years and find no discontinuity in prison outcomes for any demographic group as shown in Figure 7 of the Appendix. Together, with smooth birth characteristics and the absence of differential migration, this provides further confidence that my results reflect the effect of delayed kindergarten entry on adult crime.

VI Results

VI.A Adult Crime

I first estimate equation 4 among all prisoners. Figure 4 displays the RDD plot of all prisoners relative to the cutoff. I find no aggregate effect on the imprisonment due to the kindergarten cutoff date. The absence of an aggregate effect is common in the crime literature, as crime rates vary greatly by demographic group with few individuals on the margin of being a criminal (e.g., Landersø, Nielsen, and Simonsen 2015; Depew and Eren 2016; McAdams 2016). I continue in the following section by examining heterogeneous effects among demographic groups.

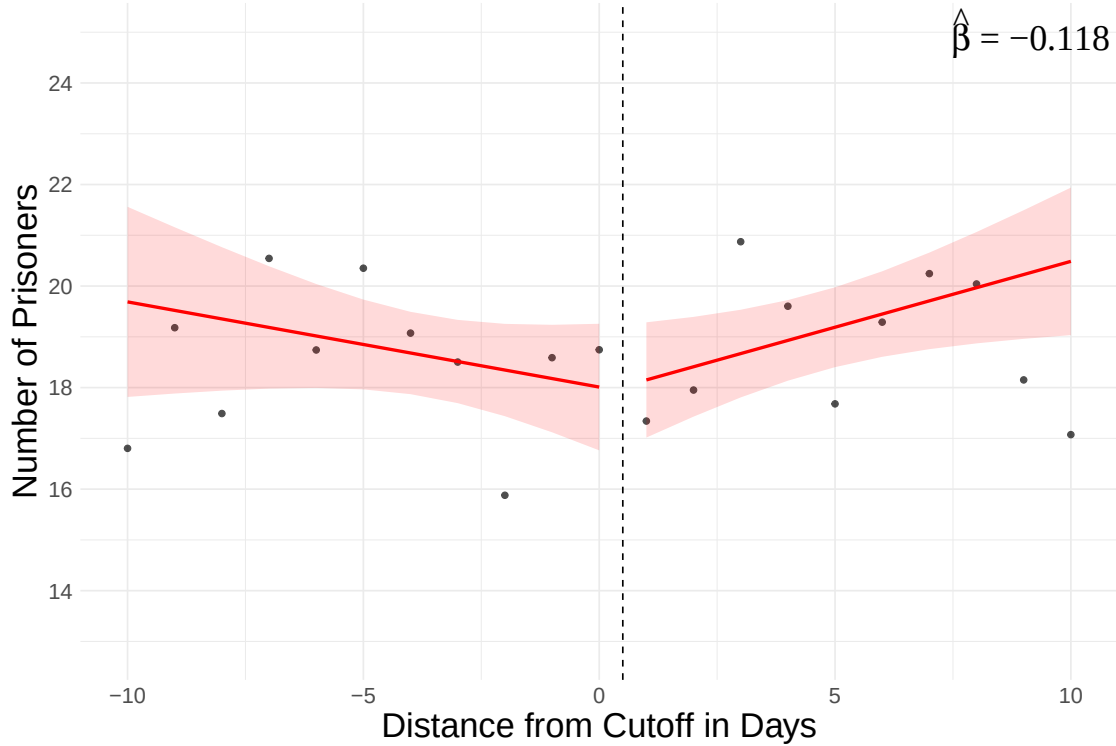
VI.B Heterogeneity By Demographic Group

I examine heterogeneity by gender, race, and gender-race pairs using my RDD on demographic subsamples. Because my unit of observation is the date of birth, splitting by demographics reduces the number of prison stays on a given date of birth but does not reduce statistical power. I report the results in Table 1 and Figure 5 displays the corresponding RD plots.

I find that delayed entry increases the number of female prisoners. The effect for females is concentrated among Whites who see an increase in the number of prisoners by 0.720 per day of birth delayed, about a 47% increase relative to the mean. My results support the notion that White women born just after the new kindergarten cutoff dates have less human capital, on average, and commit more crimes than those born just before the cutoff (Dobkin and Ferreira 2010; Kaplan, Spenkuch, and Tuttle 2025).

The absence of an effect on males is consistent with the theoretical predictions in Section III, and reflects heterogeneity in the relationship between education and crime by gender. All estimates should be interpreted as lower bounds of the local average treatment effect (LATE), as academic redshirting, migration, and the absence of birthplace data cause birth date to be an imperfect measure of school start date.

Figure 4: Regression Discontinuity Plot of All Prisoners



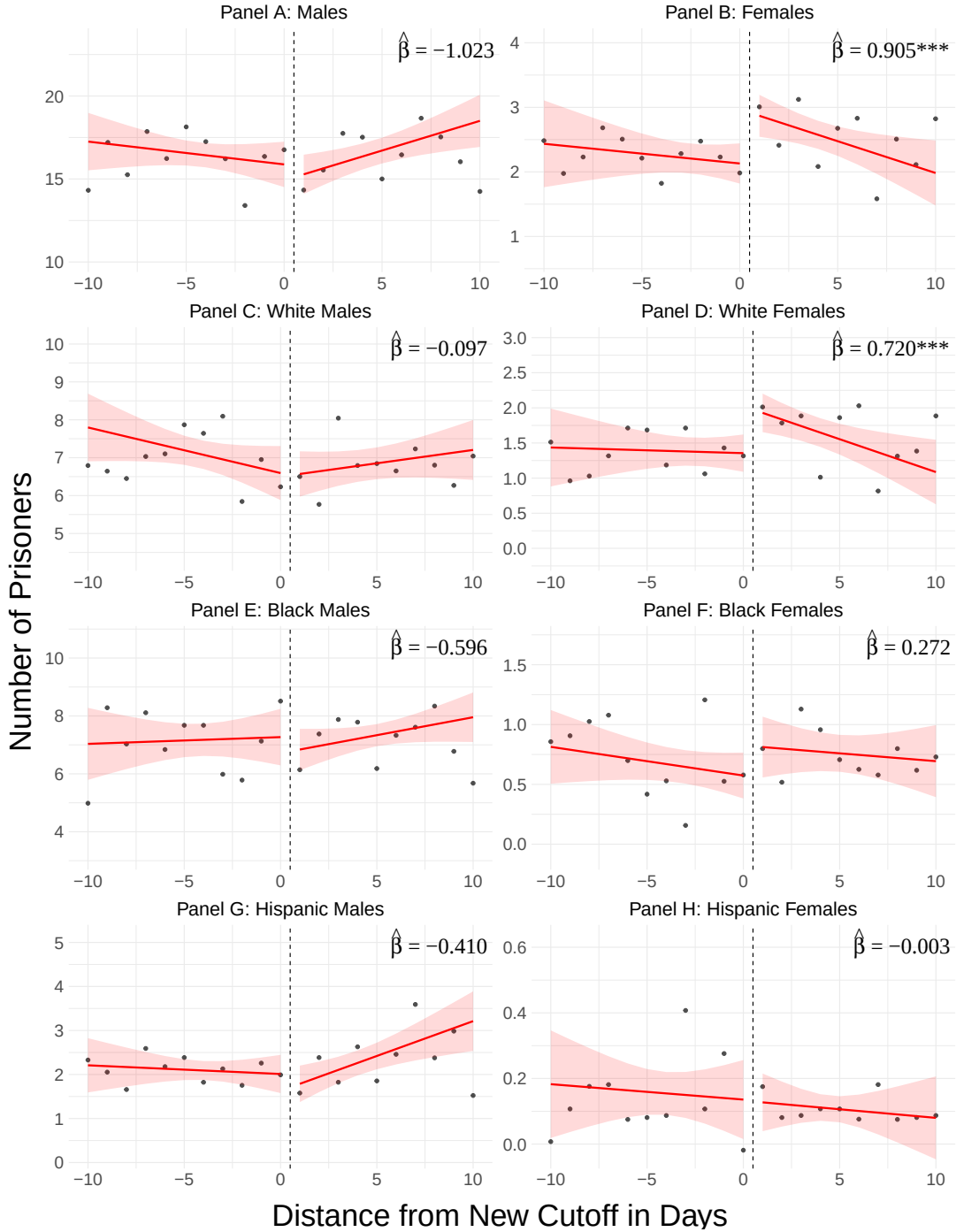
Note: The horizontal axis is the number of days to the new kindergarten entry cutoff and is normalized to zero. Individuals with a positive distance were delayed school entry due to the policy change. Zero corresponds to either December 1, 1975, November 1, 1976, October 1, 1977, September 1, 1978, or September 1, 1979 in Florida and either November 1, 1981, October 1, 1982, September 1, 1983, September 1, 1984, or September 1, 1985 in Illinois. Points show residualized bin means after partialing out day-of-week and birth year fixed effects via the Frisch-Waugh theorem (Lovell 1963). The shaded area represents 95 percent point-wise confidence intervals around the fitted regression function, based on heteroskedasticity robust standard errors clustered at the month-day level.

Using first-stage estimates on the effect of delayed entry on school attainment, I calculate back-of-the-envelope local average treatment effects of education on lifetime crime. Using the first-stage estimate from Dobkin and Ferreira (2010), I find that a one percentage point increase in high school attainment decreases the crime rate of White women by over 10%. Using the first-stage estimate from Kaplan, Spenkuch, and Tuttle (2025), increasing education by one year decreases the probability that a White woman commits crime by 5%–15%.

VI.C Age Effects and Crime Composition

I generally find no effect on the age at which a prisoner first ends up in prison, as reported in Table 12 in the Appendix. Since IDOC data begin in 2005, I do not observe prison stays before age 24 for the earliest affected cohorts, truncating the distribution and pushing my observed average age of first entry upward.

Figure 5: Prisoner Regression Discontinuity Plots by Demographic Group



Note: The horizontal axis is the number of days to the new kindergarten entry cutoff and is normalized to zero. Individuals with a positive distance were delayed school entry due to the policy change. Zero corresponds to either December 1, 1975, November 1, 1976, October 1, 1977, September 1, 1978, or September 1, 1979 in Florida and either November 1, 1981, October 1, 1982, September 1, 1983, September 1, 1984, or September 1, 1985 in Illinois. Points show residualized bin means after partialing out day-of-week and birth year fixed effects via the Frisch-Waugh theorem (Lovell 1963). The shaded area represents 95 percent point-wise confidence intervals around the fitted regression function, based on heteroskedasticity robust standard errors clustered at the month-day level.

Table 1: Regression Discontinuity Results by Race and Gender

	All	White	Black	Hispanic
<i>Panel A: Males and Females</i>				
Delayed	-0.118 (0.978) {18.662}	0.623 (0.592) {8.381}	-0.324 (0.716) {7.890}	-0.413 (0.339) {2.267}
<i>Panel B: Males Only</i>				
Delayed	-1.023 (1.021) {16.267}	-0.097 (0.552) {6.867}	-0.596 (0.686) {7.186}	-0.410 (0.338) {2.133}
<i>Panel C: Females Only</i>				
Delayed	0.905*** (0.273) {2.395}	0.720*** (0.231) {1.514}	0.272 (0.174) {0.705}	-0.003 (0.089) {0.133}
Bandwidth (Days)	10	10	10	10
<i>N</i> (Days of Birth)	210	210	210	210

Note: Sample means, within the bandwidth, are reported in the braces. The dependent variable is the number of prisoners born a given number of days from the new cutoff, for each demographic group, in Florida and Illinois. $Delayed_{dts}$ is a dummy indicating that people born in that state on that date of birth had their kindergarten start date moved because of the policy change. The estimated coefficient, β , from equation 4 is reported along with its standard error, clustered at the month-day level, (in parentheses) and outcome average {in braces}. * significant at 10%, ** at 5%, *** at 1%.

Similarly, I find no effect on aggregate crime composition as reported in Table 13 of the Appendix.

Table 2 estimates the effects by crime type for white women. The increase in imprisonment among delayed white women is concentrated in property crimes with no effect on other types. Property crimes represent financially motivated crimes, these being the crimes most affected by changes in legal earnings (Gould, Weinberg, and Mustard 2002; Machin and Meghir 2004). An effect on only property crimes is consistent with the interpretation that delayed entry reduces wages and shifts White women toward the illegal labor market (Lochner and Moretti 2004; Cano-Urbina and Lochner 2019). While small means by crime type limit precision, the pattern corroborates the broader mechanism in Section III and helps to explain the results.

Table 2: White Females Crime Category Regression Discontinuity Results

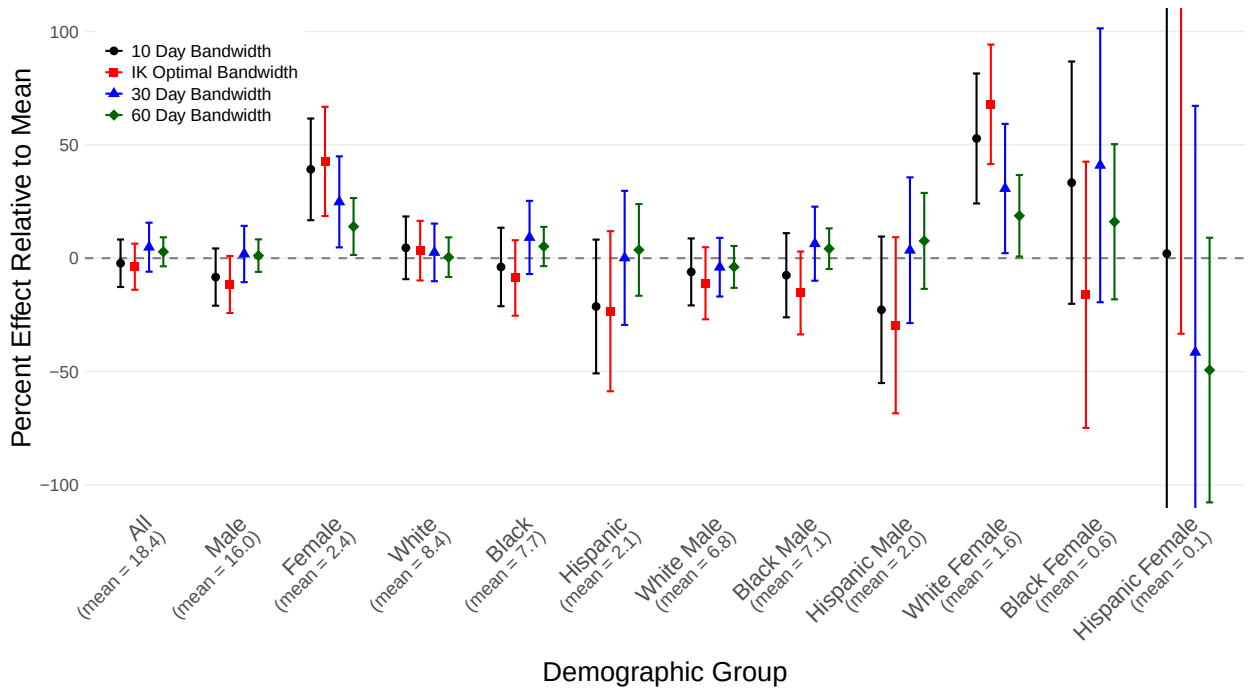
	Violent	Drug	Property	Sex	Other
Delayed	0.105 (0.101) {0.110}	0.009 (0.148) {0.333}	0.266** (0.124) {0.295}	0.041 (0.043) {0.029}	0.068 (0.049) {0.014}
Bandwidth (Days)	10	10	10	10	10
<i>N</i> (Days of Birth)	210	210	210	210	210

Note: Sample is restricted to only white females. The dependent variable is the number of prisoners born on a day a given number of days from the new cutoff, who was convicted of the given crime type, in Florida and Illinois. $Delayed$ is a dummy indicating that people born in that state on that date of birth had their kindergarten start date moved because of the policy change. The estimated coefficient, β , from equation 4 is reported along with its standard error, clustered at the month-day level, (in parentheses) and outcome average {in braces}. * significant at 10%, ** at 5%, *** at 1%.

VI.D Robustness

I check the robustness of my results to varying bandwidths with results in Table 3 and Figure 6. I also check against all possible clustering methods and polynomial orders with the results reported in Table 10 and Table 11 of the Appendix, respectively.²⁴ I find my estimates to be robust to different bandwidths, clustering methods, and polynomial orders. I find a consistent effect on females that is concentrated among Whites. Across all specifications, the smallest effect I find is a 16% increase in White female crime due to delayed school entry.

Figure 6: Robustness Checks by Bandwidth Selection



Note: The bars represent 95 percent confidence intervals, based on standard errors clustered at the month-day level. All estimates in this figure use a triangular kernel.

VII Conclusion

This paper is the first to exploit quasi-random variation in kindergarten entry cutoff dates to measure their effect on lifetime crime. I do so using policy changes in Florida and Illinois during the 1980s, which

²⁴Table 11 includes combinations of ten day and IK optimal bandwidths with all levels of clustering and a triangular kernel. Month-day or state-month are the only levels which have more than thirty clusters. I also present results with robust standard errors but no clustering. Table 11 includes results at all possible polynomial orders. A ten day bandwidth allows for identification of up to a cubic polynomial. A quartic polynomial results in a rank-deficient model due to only twenty-one discrete values of the running variable.

occurred after conception. I pair administrative prison data from each state, which include exact dates of birth, with a regression discontinuity design where the running variable is the distance between the date of birth and the cutoff date.

I find evidence of an effect on women concentrated among White females, with delayed school entry increasing the number of White women who ever go to prison on delayed dates of birth by 0.720 prisoners per day. Compared to the mean, this represents an increase of at least 16%, even by my most modest estimate. Using NVSS and ACS data, I show this reflects an increase in the crime rate rather than changes in births or migration.

The concentration of effects among white women is not unprecedented and follows directly from the theory model in Section III (Depew and Eren 2016; Cano-Urbina and Lochner 2019). The model predicts larger wage gaps, $w_i(h_i^H) - w_i(h_i^L)$, between levels of human capital to result in a larger effect on crime concentrated in property offenses, holding the effect on education constant. The literature suggests women have roughly 15% higher returns to education, on average, compared to men (e.g., Neumark 1988; Gronau 1988; Card 1999; Dougherty 2005).²⁵ These returns are highest during high school (e.g., Ashenfelter and Krueger 1994; Behrman, Rosenzweig, and Taubman 1994; Dougherty 2005; Blau and Kahn 2017). The same is true of the relationship between education and crime, which is strongest at the high school margin (Lochner and Moretti 2004).

Marriage market returns reinforce this pattern for White women specifically. Becker (1991) predicts that increases in education raise the education and earnings of a spouse through assortative mating. Empirical evidence shows marriage market returns to be substantial for White women but mostly absent for men and non-White women (Behrman, Rosenzweig, and Taubman 1994; Chiappori, Iyigun, and Weiss 2009). Marriage returns are also highest at lower education levels, exactly where compulsory schooling binds (Lefgren and McIntyre 2006; McCrary and Royer 2011). Such gains raise the opportunity cost of criminal activity in ways that do not operate through one's own labor market. The combination of these channels widens the opportunity cost differential for White women relative to all other groups.²⁶

The null result for men is consistent with the model and literature, but requires additional explanation. First, lower returns to education for men result in a lower wage gap thus reducing the effect (e.g., Neumark 1988; Gronau 1988; Card 1999; Dougherty 2005). Secondly, male criminal behavior is driven by a wider set of factors than that of females. Female crime is believed to be, average, more financially motivated whereas males' include anger and social group motivations (Steffensmeier and Allan 1996). This implies the labor

²⁵Shenhav and Sandler (2026) further examine the multidimensional gains to education for women using a similar design. They find education to simultaneously effect employment, earnings, occupational status, and working conditions.

²⁶The concentration among White women is further consistent with evidence that access to formal schooling at age five reduces the probability of dropping out and institutionalization among only Whites (Cascio 2009).

market channel to be only one of many dimensions for men and concentrates effects for females.

My estimates for White women are modestly higher than those of the literature. Past literature finds delayed enrollment to increase adult incarceration by over 3% and felony convictions, aged 17-19, by 14% (Cook and Kang 2016; McAdams 2016). These differences can be explained by my paper using a much closer control group than previously in the literature. Cook and Kang (2016) use a sixty day bandwidth, while McAdams (2016) does a difference-in-differences using quarters of birth on either side of the cutoff. My preferred specification includes only those born within ten days of the cutoff. My narrower bandwidth makes both observable and unobservable factors more likely to be indistinguishable across the cutoff due to the seasonality of births and potential for parents to plan births (Buckles and Hungerman 2013; Clarke, Orefice, and Quintana-Domeque 2019). Additionally, my data include the entire prison population in Florida and Illinois which prevents selection bias of prisoners on those serving the most time or who committed crime most recently to the survey period.

Scaling my estimates by the first-stage effects of delayed entry on educational attainment from Dobkin and Ferreira (2010) and Kaplan, Spenkuch, and Tuttle (2025), I find that a one percentage point increase in high school attainment decreases White female crime by over 10%, and that one additional year of schooling decreases the probability that a White woman commits crime by over 5%. These magnitudes align closely with the broader human capital and crime literature, where Lochner and Moretti (2004) find that a one-year increase in average schooling reduces state-level arrest rates by approximately 11%. Together, these results highlight the lifelong consequences of kindergarten entry cutoff dates for female crime and provide new evidence on the effect of delayed school entry on lifetime incarceration.

Table 3: Results by Demographic Group and Bandwidth

	(1)	(2)	(3)	(4)	(5)
All	-0.118 (0.978)	-0.296 (0.961)	0.510 (0.768)	0.725 (0.734)	0.518 (0.558)
Males	-1.023 (1.021)	-1.256 (1.053)	-0.028 (0.753)	0.191 (0.703)	0.196 (0.552)
Females	0.905*** (0.273)	0.950*** (0.298)	0.538*** (0.215)	0.534** (0.220)	0.322** (0.149)
Whites	0.623 (0.594)	0.911 (0.636)	0.089 (0.454)	0.131 (0.496)	0.033 (0.354)
Blacks	-0.324 (0.716)	-0.785 (0.676)	0.187 (0.486)	0.380 (0.429)	0.408 (0.334)
Hispanics	-0.413 (0.339)	-0.184 (0.432)	0.233 (0.249)	0.179 (0.272)	0.071 (0.218)
White Males	-0.097 (0.552)	-0.056 (0.612)	-0.360 (0.387)	-0.403 (0.411)	-0.242 (0.302)
Black Males	-0.596 (0.686)	-1.276 (0.598)	0.047 (0.468)	0.291 (0.421)	0.301 (0.317)
Hispanic Males	-0.410 (0.338)	-0.272 (0.414)	0.284 (0.256)	0.267 (0.264)	0.141 (0.212)
White Females	0.720*** (0.231)	0.938*** (0.235)	0.449*** (0.189)	0.533*** (0.196)	0.275* (0.142)
Black Females	0.272 (0.192)	-0.041 (0.211)	0.140 (0.137)	0.089 (0.146)	0.107 (0.108)
Hispanic Females	-0.003 (0.023)	0.173 (0.110)	-0.051 (0.057)	-0.088 (0.063)	0.018 (0.020)
IK Optimal?		Yes			
Cluster Level	Month-Day	Month-Day	Month-Day	Month-Day	Month-Day
Bandwidth (Days)	10	4–9	29	30	60
N (Days of Birth)	210	90–190	590	610	1,089

Note: The dependent variable is the number of prisoners born a given number of days from the new cutoff, for each demographic group, in Florida and Illinois. The coefficient of Delayed_{dts} is reported being the effect of delayed kindergarten entry on the number of prisoners with that date of birth. IK optimal represents optimal bandwidths suggested by Imbens and Kalyanaraman (2012) and rounded to the nearest whole number. Optimal bandwidths are reported in Table 9 of the Appendix. When the bandwidth is greater than twenty-nine days, those born in 1983 are excluded to account for potential confoundedness created by Medicaid expansion to kids. The estimated coefficient, β , from equation 4 is reported along with its standard error, clustered at the month-day level, (in parentheses). * significant at 10%, ** at 5%, *** at 1%.

VIII Appendix

Table 4: Kindergarten Cutoff Birthday by State and Year

State	Pre-1980	1980	1981	1982	1983	1984-1985	1986	1987	1988	Post-1988
Florida	Jan 1	Dec 1	Nov 1	Oct 1	Sept 1	Sept 1	Sept 1	Sept 1	Sept 1	Sept 1
Illinois	Dec 1	Dec 1	Dec 1	Dec 1	Dec 1	Dec 1	Nov 1	Oct 1	Sept 1	Sept 1

Note: Based on information from Florida Senate Committee on Education Innovation (1999), Florida Legislature (1979), and Education Week (1987). Those delayed by these policies include individuals born between the following dates (inclusive) in Florida: December 2, 1975 – January 1, 1976, November 2, 1976 – January 1, 1977, October 2, 1977 – January 1, 1978, September 2, 1978 – January 1 1979, and September 2, 1979 – January 1, 1980. The following were delayed (inclusive) in Illinois: November 2 – December 1, 1981, October 2 – December 1, 1982, September 2 – December 1, 1983, September 2 – December 1, 1984, and September 2 – December 1, 1985.

Table 5: Prison Admissions Data Availability by Age

Birthday Cohort	Age									
<i>Panel A: Florida</i>	18	19	20	21	22	23	24	25	26+	
Dec 2, 1975 – Dec 1, 1976	X	X	X	X	X	X	X	X	X	
Dec 2, 1976 – Nov 1, 1977	X	X	X	X	X	X	X	X	X	
Oct 2, 1977 – Sept 1, 1978	X	X	X	X	X	X	X	X	X	
Sept 2, 1978 – Sept 1, 1979	X	X	X	X	X	X	X	X	X	
Sept 2, 1979 – Sept 1, 1980	X	X	X	X	X	X	X	X	X	
<i>Panel B: Illinois</i>	18	19	20	21	22	23	24	25	26+	
Dec 2, 1979 – Dec 1, 1980								X	X	
Dec 2, 1980 – Nov 1, 1981							X	X	X	
Nov 2, 1981 – Oct 1, 1982						X	X	X	X	
Oct 2, 1982 – Sept 1, 1983					X	X	X	X	X	
Sept 2, 1983 – Sept 1, 1984				X	X	X	X	X	X	
Sept 2, 1984 – Sept 1, 1985			X	X	X	X	X	X	X	
Sept 2, 1985 – Sept 1, 1986		X	X	X	X	X	X	X	X	

Note: Florida Offender Based Information System (FOBIS) prison admissions data is available from 1981 onward while Illinois Department of Corrections (IDOC) data is only available beginning in 2005. As a result, the number of observed years of criminal history increases for later cohorts. I observe admissions data starting at age 24 for the first cohort affected by the policy, and gain an additional year for each subsequent cohort.

Table 6: Prisoner Summary Statistics

Variable	Entire Affected Cohorts					Affected Cohorts w/in Bandwidth				
	Mean	SD	Min	Max	Total	Mean	SD	Min	Max	Total
Florida Prisoners	14.908	4.643	3	34	74.539	16.045	4.574	5	30	80.226
Illinois Prisoners	19.697	5.438	4	41	98.480	20.981	5.183	10	37	104.905
Males	15.026	5.293	2	38	150.268	15.979	5.170	4	34	159.791
Females	2.276	1.603	0	11	22.751	2.471	1.631	0	10	24.710
Whites	7.807	3.201	0	24	78.065	8.255	3.123	2	17	82.552
Blacks	7.349	3.609	0	26	73.488	7.884	3.482	1	19	78.840
Hispanics	2.028	1.796	0	11	20.278	2.176	1.805	0	8	21.763
White Males	6.311	2.770	0	19	63.116	6.643	2.649	1	15	66.427
Black Males	6.725	3.488	0	21	67.249	7.202	3.337	0	18	72.019
Hispanic Males	1.895	1.746	0	10	18.942	2.030	1.738	0	8	20.302
White Females	1.495	1.282	0	8	14.949	1.613	1.316	0	6	16.125
Black Females	0.624	0.811	0	6	6.239	0.682	0.832	0	5	6.821
Hispanic Females	0.134	0.371	0	3	1.336	0.146	0.380	0	2	1.462
Age	31.280	4.331	18.146	47.961	312.803	31.274	4.224	22.301	45.032	312.744
Violent Crimes	3.238	3.772	0	18	32.378	3.990	4.027	0	16	34.408
Drug Crimes	3.675	4.238	0	20	36.745	4.038	4.341	0	16	37.726
Property Crimes	1.988	2.529	0	14	19.875	2.510	2.652	0	12	20.046
Sex Crimes	0.718	1.130	0	7	7.185	0.881	1.209	0	7	7.865
Other Crimes	0.226	0.538	0	5	2.253	0.205	0.518	0	3	2.088

Note: Table includes the summary stats of observations at the date of birth level for cohorts in Florida and Illinois affected by the change in the kindergarten entry cutoff date after conception. This includes those born in Florida from 1975 to 1979 and in Illinois from 1981 to 1985. The right panel includes only dates of births within ten days of the new cutoff entry day. The demographic groups listed corresponds to prisoners of that demographic group observed in the combined FOBIS and IDOC data. Similarly, types of crimes listed correspond to prisoners who committed a crime of that category.

Table 7: Natality Regression Discontinuity Results

Outcome	Delayed	Average	Bandwidth (Days)	N (Days of Birth)
Births	-0.278 (21.694)	889.252	10	210
Male Births	-11.924 (11.761)	455.429	10	210
Female Births	-15.855 (10.978)	433.824	10	210
White Births	-22.872 (19.979)	665.105	10	210
Black Births	-5.805 (5.079)	210.129	10	210
White Male Births	-8.274 (10.828)	341.729	10	210
Black Male Births	-4.620 (3.126)	106.376	10	210
White Female Births	-14.598 (10.181)	323.376	10	210
Black Female Births	1.185 (3.979)	103.752	10	210
Birth Weight	4.219 (7.483)	3,318.204	10	210
Mother's Age	0.022 (0.078)	25.068	10	210
Father's Age	-0.075 (0.092)	28.344	10	210
Mother's Education	-0.048 (0.034)	12.152	10	210
Father's Education	-0.049 (0.033)	12.722	10	210
% Male	0.003 (0.004)	0.512	10	210
% Hospital Births	-0.001 (0.001)	0.990	10	210
Prenatal Visits	-0.027 (0.068)	10.167	10	210

Note: The dependent variable is indicated in the outcome column. *Delayed* is a dummy indicating that people born in that state on that date of birth had their kindergarten start date moved because of the policy change. The estimated coefficient, α , from equation 5 is reported along with its standard error, clustered at the month-day level, (in parentheses). Source: National Center for Health Statistics (1975-1985). * significant at 10%, ** at 5%, *** at 1%.

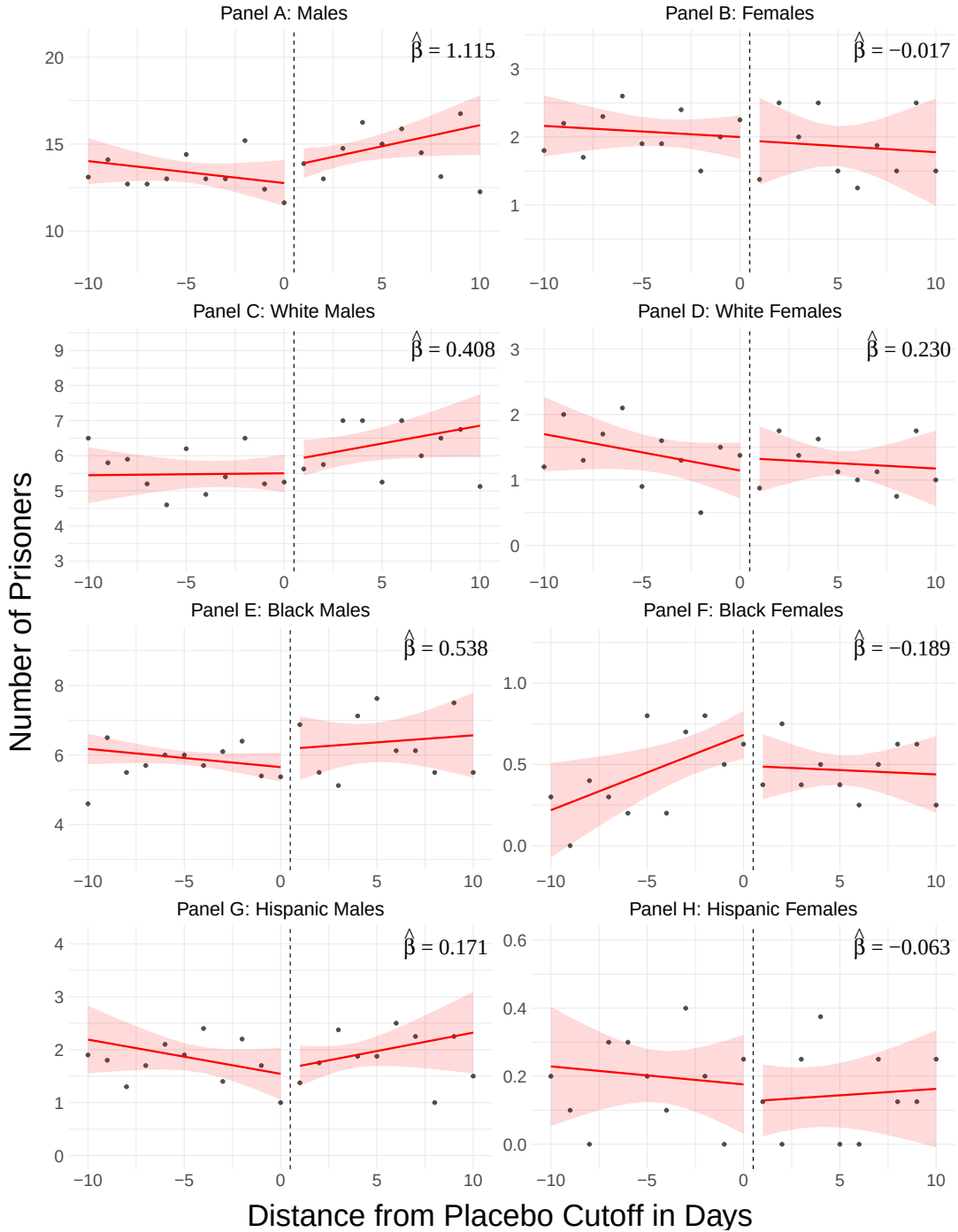
Table 8: Moving by Quarter of Birth (1977 and 1982 Birth Cohorts)

	Moving Out				Moving In			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Delayed	-0.006 (0.008)				0.000 (0.001)			
Delayed×Male		-0.018 (0.012)				-0.002 (0.002)		
Delayed×Female		0.007 (0.012)				-0.002 (0.003)		
Delayed×White			-0.012 (0.010)				-0.003 (0.002)	
Delayed×Black			-0.010 (0.019)				0.003 (0.003)	
Delayed×Hispanic			0.028 (0.024)				0.001 (0.005)	
Delayed×Male×White				-0.033** (0.013)				-0.003 (0.003)
Delayed×Male×Black				0.005 (0.024)				0.001 (0.003)
Delayed×Male×Hispanic				0.022 (0.030)				-0.002 (0.006)
Delayed×Female×White				0.010 (0.014)				-0.004 (0.004)
Delayed×Female×Black				-0.024 (0.024)				0.004 (0.004)
Delayed×Female×Hispanic				0.035 (0.030)				0.003 (0.007)
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster Level	Household	Household	Household	Household	Household	Household	Household	Household
<i>N</i>	2,648,081	2,648,081	2,648,081	2,648,081	5,816,268	5,816,268	5,816,268	5,816,268

Note: Standard errors in parentheses, clustered at the household level. The moving out sample is restricted to those born in the third and fourth quarters of 1977, in Florida, and 1982, in Illinois. These individuals were affected by the policy change after conception and the new cutoff date (10/1) aligned with a quarter of birth, the most specific level of birth date information available in the ACS. The moving in sample is restricted to those born in the third and fourth quarters of 1977 and 1982 and reside in Florida or Illinois at the time of survey. The dependent variable is an indicator equal to one if the individual no longer lives in their state of birth. *Delayed* is a dummy indicating that individual was delayed school entry due to the policy change.

* significant at 10%, ** at 5%, *** at 1%.

Figure 7: Placebo Regression Discontinuity Plots



Note: The horizontal axis is the number of days to the new kindergarten entry cutoff and is normalized to zero. Individuals with a positive distance were delayed school entry due to the policy change. Zero corresponds to May 1 in either the year 1975, 1976, 1977, 1978, or 1979 in Florida and in either 1981, 1982, 1983, 1984, or 1985 in Illinois. Points show residualized bin means after partialing out day-of-week and birth year fixed effects via the Frisch-Waugh theorem (Lovell 1963). The shaded area represents 95 percent point-wise confidence intervals around the fitted regression function, based on heteroskedasticity robust standard errors clustered at the month-day level.

Table 9: IK Optimal Bandwidths

Outcome Variable	Bandwidth
All Prisoners	9
Male Prisoners	8
Female Prisoners	5
White Prisoners	7
Black Prisoners	8
Hispanic Prisoners	6
White Male Prisoners	6
Black Male Prisoners	7
Hispanic Male Prisoners	6
White Female Prisoners	5
Black Female Prisoners	5
Hispanic Female Prisoners	4
All Prisoners' Age	8
Male Prisoners' Age	8
Female Prisoners' Age	8
White Prisoners' Age	7
Black Prisoners' Age	9
Hispanic Prisoners' Age	9
White Male Prisoners' Age	7
Black Male Prisoners' Age	9
Hispanic Male Prisoners' Age	9
White Female Prisoners' Age	8
Black Female Prisoners' Age	11
Hispanic Female Prisoners' Age	12
Violent Crimes	8
Property Crimes	7
Drug Crimes	7
Sex Crimes	5
Other Crimes	4

Note: Bandwidths represent the optimal bandwidth suggested by Imbens and Kalyanaraman (2012), for each outcome, which have been rounded to the nearest whole number.

Table 10: Regression Discontinuity Results by Clustering Method

	(1)	(2)	(3)	(4)	(5)	(6)
All	-0.118 (0.978)	-0.296 (0.961)	-0.118 (1.009)	-0.296 (0.990)	-0.417 (0.969)	-0.808 (0.978)
Males	-1.023 (1.021)	-1.256 (1.015)	-1.023 (0.918)	-1.256 (0.922)	-1.023 (1.044)	-1.256* (1.106)
Females	0.905*** (0.273)	0.980*** (0.288)	0.905*** (0.306)	0.980*** (0.363)	0.905*** (0.303)	0.980*** (0.375)
Whites	0.623 (0.594)	0.911 (0.634)	0.623 (0.531)	0.911 (0.921)	0.623 (0.707)	0.911 (0.877)
Blacks	-0.324 (0.716)	-0.785 (0.676)	-0.324 (0.747)	-0.786 (0.734)	-0.324 (0.623)	-0.785 (0.608)
Hispanics	-0.413 (0.339)	-0.184 (0.432)	-0.413* (0.241)	-0.184* (0.629)	-0.413 (0.322)	-0.184 (0.464)
White Males	-0.097 (0.552)	0.056 (0.619)	-0.097 (0.376)	0.056 (1.138)	-0.097 (0.697)	0.056 (0.972)
Black Males	-0.596 (0.686)	-1.276 (0.598)	-0.596 (0.703)	-1.276 (0.688)	-0.596 (0.595)	-1.276** (0.567)
Hispanic Males	-0.410 (0.338)	-0.272 (0.414)	-0.410* (0.240)	-0.272 (0.531)	-0.410 (0.313)	-0.272 (0.444)
White Females	0.720*** (0.231)	0.720*** (0.235)	0.720*** (0.290)	0.938*** (0.321)	0.720*** (0.247)	0.720*** (0.285)
Black Females	0.272 (0.174)	-0.041 (0.211)	0.272 (0.169)	-0.041 (0.215)	0.272* (0.157)	-0.041 (0.186)
Hispanic Females	-0.003 (0.089)	0.173 (0.110)	-0.003 (0.017)	0.173 (0.187)	-0.003 (0.107)	0.173 (0.152)
IK Optimal?		Yes		Yes		Yes
Cluster Level	Month-Day	Month-Day	State-Month	State-Month	None	None
Bandwidth (Days)	10	4–9	10	4–9	10	4–9
N (Days of Birth)	210	90–190	210	90–190	210	90–190

Note: The dependent variable is the number of prisoners born a given number of days from the new cutoff, for each demographic group, in Florida and Illinois. The coefficient of Delayed_{dts} is reported being the effect of delayed kindergarten entry on the number of prisoners with that date of birth. IK optimal represents optimal bandwidths suggested by Imbens and Kalyanaraman (2012) and rounded to the nearest whole number. Optimal bandwidths are reported in Table 9 of the Appendix. When the bandwidth is greater than twenty-nine days, those born in 1983 are excluded to account for potential confoundedness created by Medicaid expansion to kids. The estimated coefficient, β , from equation 4 is reported along with its standard error (in parentheses). * significant at 10%, ** at 5%, *** at 1%.

Table 11: Regression Discontinuity Results by Polynomial Order

	(1)	(2)	(3)
All	-0.118 (0.978)	-1.761 (1.137)	-3.935** (1.852)
Males	-1.023 (1.021)	-2.818** (1.215)	-5.028*** (1.724)
Females	0.905*** (0.273)	1.057*** (0.335)	1.093* (0.604)
Whites	0.623 (0.594)	0.714 (0.887)	0.355 (1.423)
Blacks	-0.324 (0.716)	-2.047*** (0.676)	-3.916*** (0.937)
Hispanics	-0.413 (0.339)	-0.398 (0.512)	-0.477 (0.659)
White Males	-0.097 (0.552)	-0.276 (0.820)	-0.714 (1.273)
Black Males	-0.596 (0.686)	-2.043*** (0.623)	-3.712*** (0.849)
Hispanic Males	-0.410 (0.338)	-0.506 (0.506)	-0.721 (0.638)
White Females	0.720*** (0.231)	0.989*** (0.304)	1.070** (0.442)
Black Females	0.272 (0.174)	-0.004 (0.269)	-0.204 (0.489)
Hispanic Females	-0.003 (0.089)	0.108 (0.119)	0.243 (0.164)
Polynomial Order	1	2	3
Cluster Level	Month-Day	Month-Day	Month-Day
Bandwidth (Days)	10	10	10
N (Days of Birth)	210	210	210

Note: Standard errors in parentheses, clustered at the month-day level. The dependent variable is the number of prisoners born a given number of days from the new cutoff, for each demographic group, in Florida and Illinois. The coefficient of Delayed_{dts} is reported being the effect of delayed kindergarten entry on the number of prisoners with that date of birth. The estimated coefficient, β , from equation 4 is reported along with its standard error (in parentheses). * significant at 10%, ** at 5%, *** at 1%.

Table 12: Age Regression Discontinuity Results by Race and Gender

	All	White	Black	Hispanic
<i>Panel A: Males and Females</i>				
Delayed	-0.620 (0.724) {30.833}	-1.173 (0.855) {32.223}	-0.232 (1.486) {28.802}	-1.662 (1.868) {28.281}
<i>Panel B: Males Only</i>				
Delayed	-1.233 (0.985) {30.313}	-1.452 (1.016) {31.721}	-1.044 (1.964) {28.119}	-0.702 (1.662) {28.070}
<i>Panel C: Females Only</i>				
Delayed	1.405 (1.139) {33.267}	0.051 (1.382) {33.596}	5.568*** (1.869) {31.132}	-12.090** (4.836) {28.965}
Cluster Level	Month-Day	Month-Day	Month-Day	Month-Day
Bandwidth (Days)	10	10	10	10
<i>N</i> (Days of Birth)	210	210	210	210

Note: The dependent variable is the average age of first offense, resulting in imprisonment, of prisoners born a given number of days from the new cutoff, for each demographic group, in Florida and Illinois. *Delayed* is a dummy indicating that people born in that state on that date of birth had their kindergarten start date moved because of the policy change. The estimated coefficient, β , from equation 4 is reported along with its standard error, clustered at the month-day level, (in parentheses) and outcome average {in braces}. * significant at 10%, ** at 5%, *** at 1%.

Table 13: Crime Category Regression Discontinuity Results

	Violent	Drug	Property	Sex	Other
Delayed	0.355 (0.502) {3.990}	-0.140 (0.468) {4.038}	-0.053 (0.428) {2.510}	-0.073 (0.150) {0.881}	0.078 (0.108) {0.205}
Cluster Level	Month-Day	Month-Day	Month-Day	Month-Day	Month-Day
Bandwidth (Days)	10	10	10	10	10
<i>N</i> (Days of Birth)	210	210	210	210	210

Note: The dependent variable is the number of prisoners born a given number of days from the new cutoff, who was convicted of the given crime type, in Florida and Illinois. *Delayed* is a dummy indicating that people born in that state on that date of birth had their kindergarten start date moved because of the policy change. The estimated coefficient, β , from equation 4 is reported along with its standard error, clustered at the month-day level, (in parentheses) and outcome average {in braces}. * significant at 10%, ** at 5%, *** at 1%.

References

- Abbott, E. and S. P. Breckinridge (1917). *Truancy and Non-Attendance in the Chicago Schools: A Study of the Social Aspects of the Compulsory Education and Child Labor Legislation of Illinois*. Chicago, IL, USA: The University of Chicago Press.
- Anderson, D. M. (2014). “In School and Out of Trouble? The Minimum Dropout Age and Juvenile Crime”. *Review of Economics and Statistics* 96.2, pp. 318–331.
- Angrist, J. D. and A. B. Krueger (1991). “Does Compulsory School Attendance Affect Schooling and Earnings?” *Quarterly Journal of Economics* 106.4, pp. 979–1014.
- Arenberg, S., S. Neller, and S. Stripling (2024). “The Impact of Youth Medicaid Eligibility on Adult Incarceration”. *American Economic Journal: Applied Economics* 16.1, pp. 121–156.
- Ashenfelter, O. and A. B. Krueger (1994). “Estimates of the Economic Returns to Schooling from a New Sample of Twins”. *American Economic Review* 84.5, pp. 1157–1173.
- Baron, E. J., J. M. Hyman, and B. N. Vasquez (2024). “Public School Funding, School Quality, and Adult Crime”. *Review of Economics and Statistics* 1.1, pp. 1–46.
- Barr, A. C., J. Eggleston, and A. A. Smith (2022). “Investing in Infants: The Lasting Effects of Cash Transfers to New Families”. *The Quarterly Journal of Economics* 137.4, pp. 2539–2583.
- Becker, G. (1968). “Crime and Punishment: An Economic Approach”. *Journal of Political Economy* 76.2, pp. 169–217.
- Becker, G. (1991). “Assortative Mating in Marriage Markets”. In: *A Treatise on the Family*. Enlarged Edition. Cambridge, MA: Harvard University Press, pp. 108–134.
- Bédard, K. and E. Dhuey (2006). “The Persistence of Early Childhood Maturity: International Evidence of Long-Run Age Effects”. *Quarterly Journal of Economics* 121.4, pp. 1437–1472.
- Behrman, J. R., M. R. Rosenzweig, and P. Taubman (1994). “Endowments and the Allocation of Schooling in the Family and in the Marriage Market: The Twins Experiment”. *Journal of Political Economy* 102.6, pp. 1131–1174.
- Bell, B., R. Costa, and S. Machin (2022). “Why Does Education Reduce Crime?” *Journal of Political Economy* 130.3, pp. 732–765.
- Black, S. E., P. J. Devereux, and K. G. Salvanes (2011). “Too Young to Leave the Nest? The Effects of School Starting Age”. *Review of Economics and Statistics* 93.2, pp. 455–467.
- Blau, F. D. and L. M. Kahn (2017). “The Gender Wage Gap: Extent, Trends, and Explanations”. *Journal of Economic Literature* 55.3, pp. 789–865.
- Brunello, G., M. Fort, and G. Weber (2009). “Changes in Compulsory Schooling, Education and the Distribution of Wages in Europe”. *Economic Journal* 119.536, pp. 516–539.
- Buckles, K. S. and D. M. Hungerman (2013). “Season of Birth and Later Outcomes: Old Questions, New Answers”. *Review of Economics and Statistics* 95.3, pp. 711–724.
- Cano-Urbina, J. and L. Lochner (2019). “The Effect of Education and School Quality on Female Crime”. *Journal of Human Capital* 13.2, pp. 188–234.
- Card, D. (1999). “The Causal Effect of Education on Earnings”. 3. Ed. by O. Ashenfelter and D. Card, pp. 1801–1863.
- Cascio, E. U. (May 2009). “Do Investments in Universal Early Education Pay Off? Long-Term Effects of Introducing Kindergartens into Public Schools”. 14951.
- Chiappori, P.-A., M. Iyigun, and Y. Weiss (2009). “Investment in Schooling and the Marriage Market”. *American Economic Review* 99.5, pp. 1689–1713.
- Clark, D. and H. Royer (2013). “The Effect of Education on Adult Mortality and Health: Evidence from Britain”. *American Economic Review* 103.6, pp. 2087–2120.
- Clarke, D., S. Oreffice, and C. Quintana-Domeque (2019). “The Demand for Season of Birth”. *Journal of Applied Econometrics* 34.5, pp. 707–723.
- Cook, P. J. and S. Kang (2016). “Birthdays, Schooling, and Crime: Evidence from a Regression Discontinuity Design”. *American Economic Journal: Applied Economics* 126.593, pp. 107–134.
- Depew, B. and O. Eren (2016). “Born on the Wrong Day? School Entry Age and Juvenile Crime”. *Journal of Urban Economics* 96, pp. 73–90.
- Dobkin, C. and F. Ferreira (2010). “Do School Entry Laws Affect Educational Attainment and Labor Market Outcomes?” *Economics of Education Review* 29.1, pp. 40–54.

- Dougherty, C. (2005). “Why Are the Returns to Schooling Higher for Women than for Men?” *Journal of Human Resources* 40.4, pp. 969–988.
- Education Commission of the States (2014). “State Kindergarten Policies”. *Education Commission of the States Reports* 2014.
- Education Week (June 1987). “Legislatures, Districts Move to Raise Age for Kindergarten”. *Education Week*.
- Ehrlich, I. (1973). “Participation in Illegitimate Activities: A Theoretical and Empirical Investigation”. *Journal of Political Economy* 81.3, pp. 521–565.
- Elder, T. E. and D. Lubotsky (2009). “Kindergarten Entrance Age and Children’s Achievement: Impacts of State Policies, Family Background, and Peers”. *Journal of Human Resources* 44.3, pp. 641–683.
- Florida House of Representatives, C. o. E. K.-1. (Mar. 4, 1999). *House Bill CS/HB 307: Schools/Kindergarten Admission. Analysis*. Bill analysis, Florida House of Representatives.
- Florida Legislature (1970). *Florida Statutes, Title XLVIII, Chapter 232: Compulsory School Attendance*.
- Florida Legislature (1979). *Summary of General Legislation: 1979*.
- Florida Legislature (2025). *Florida Statute 1003.26*.
- Florida Senate Committee on Education Innovation (1999). “Bill Analysis and Economic Impact Statement: House Bill 307”. *Florida Senate Bill Analyses*.
- Gould, E. D., B. A. Weinberg, and D. B. Mustard (2002). “Crime Rates and Local Labor Market Opportunities in the United States: 1979–1997”. *The Review of Economics and Statistics* 84.1, pp. 45–61.
- Gronau, R. (1988). “Sex-Related Wage Differentials and Women’s Interrupted Labor Careers—The Chicken or the Egg”. *Journal of Labor Economics* 6.3, pp. 277–301.
- Hjalmarsson, R., H. Holmlund, and M. J. Lindquist (2015). “The Effect of Education on Criminal Convictions and Incarceration: Causal Evidence from Micro-data”. *Economic Journal* 125.587, pp. 1290–1326.
- Illinois General Assembly (1970). “School Code of Illinois: Compulsory Attendance Law (105 ILCS 5/26-1)”. *State of Illinois Legislative Documents*. No amendments raising the dropout age from 1970 to 1999.
- Illinois State Board of Education (1983). “Compulsory Attendance Mandate Report and Preliminary Recommendations: The Age of Leaving School”.
- Imbens, G. W. and K. Kalyanaraman (2012). “Optimal Bandwidth Choice for the Regression Discontinuity Estimator”. *Review of Economic Studies* 79.3, pp. 933–959.
- Kaplan, E., J. L. Spenkuch, and C. Tuttle (2025). *From the Classroom to the Ballot Box: Turnout and Partisan Consequences of Education*. Working Paper 34355. National Bureau of Economic Research.
- Kirchmaier, T. (2024). “Commuting for Crime”. *The Economic Journal* 134.659, pp. 1173–1198.
- Knapp, D. et al. (2025). “Gateway Policy Explorer: USA Compulsory Schooling Policy Details, 1900–2024”.
- Landersø, R., H. S. Nielsen, and M. Simonsen (2015). “School Starting Age and the Crime-Age Profile”. *Economic Journal* 127.602, pp. 1096–1118.
- Lefgren, L. and F. McIntyre (2006). “The Relationship between Women’s Education and Marriage Outcomes”. *Journal of Labor Economics* 24.4, pp. 787–830.
- Lleras-Muney, A. (2005). “The Relationship Between Education and Adult Mortality in the United States”. *The Review of Economic Studies* 72.1, pp. 189–221.
- Lochner, L. (2004). “Education, Work, and Crime: A Human Capital Approach”. *International Economic Review* 45.3, pp. 811–843.
- Lochner, L. and E. Moretti (2004). “The Effect of Education on Crime: Evidence from Prison Inmates, Arrests, and Self-Reports”. *American Economic Review* 94.1, pp. 155–189.
- Lovell, M. C. (1963). “Seasonal Adjustment of Economic Time Series and Multiple Regression Analysis”. *Journal of the American Statistical Association* 58.304, pp. 993–1010.
- Lubotsky, D. and R. Kaestner (2016). “Do ‘Skills Beget Skills’? Evidence on the Effect of Kindergarten Entrance Age on the Evolution of Cognitive and Non-cognitive Skill Gaps in Childhood”. *Economics of Education Review* 53, pp. 194–206.
- Machin, S. and C. Meghir (2004). “Crime and Economic Incentives”. *Journal of Human Resources* 39.4, pp. 958–979.
- McAdams, J. M. (2016). “The Effect of School Starting Age Policy on Crime: Evidence from U.S. Microdata”. *Economics of Education Review* 54, pp. 227–241.
- McCrary, J. and H. Royer (2011). “The Effect of Female Education on Fertility and Infant Health: Evidence from School Entry Policies Using Exact Date of Birth”. *American Economic Review* 101.1, pp. 158–195.

- National Center for Education Statistics (2016). *Enrollment of 3-, 4-, and 5-year-old Children in Preprimary Programs, by Age of Child, Level of Program, Control of Program, and Attendance Status: Selected Years, 1970 Through 2015*. Digest of Education Statistics 2016, Table 202.10.
- National Center for Health Statistics (1975-1985). *Nativity Data Files, 1975-1986*. Accessed via the National Bureau of Economic Research.
- Neumark, D. (1988). "Employers' Discriminatory Behavior and the Estimation of Wage Discrimination". *Journal of Human Resources* 23.3, pp. 279-295.
- Oreopoulos, P. (2006). "Estimating Average and Local Average Treatment Effects of Education when Compulsory Schooling Laws Really Matter". *American Economic Review* 96.1, pp. 152-175.
- Rossin-Slater, M., A. Sabety, and A. Wu (Jan. 2026). *The Impact of Preschool Entry Age on Children's Behavioral and Developmental Health in Medicaid*. Working Paper 34677. National Bureau of Economic Research.
- Shenhav, N. and D. H. Sandler (Jan. 2026). *Life-Cycle Effects of Women's Education on their Careers and Children*. NBER Working Paper 34767. National Bureau of Economic Research.
- Shepard, A. K. (2024). "Arrested Development: Relative School Entry Age and Arrests during the Teenage and Young Adult Years". *Education Economics* 32.3, pp. 275-297.
- Steffensmeier, D. and E. Allan (1996). "Gender and Crime: Toward a Gendered Theory of Female Offending". *Annual Review of Sociology* 22, pp. 459-487.
- U.S. Department of Education (2024). *Illinois School Discipline Laws and Regulations*.
- Whaley, M. (1985). "The Status of Kindergarten: A Survey of the States". *Illinois State Board of Education, Department of Planning, Research and Evaluation*.